

BrickAtlas Hierarchy-2: Complete Supplementary Index

Development Manuscript
Author metadata intentionally omitted

1. Scope and Version

This supplement indexes the current Hierarchy-2 release: 40 layouts, 35 families per layout, and 1,400 questions. It accompanies the machine-readable public inputs, answers, quality records, and original website 3D interface. It replaces the previous manuscript’s appendix as the current entry point. Older Mechanism-1, Hierarchy-1, and retired grid experiments remain in their versioned data directories and Git history; their scores are not pooled into this release.

2. Reading the Contracts

Each task has a source ID, difficulty stratum, task layer, family, capability, output format, evidence class, and public input. Single choices return `choiceId`; set-valued choices return `choiceIds`; operations return `actionIds`; schedules return a job-to-start-time map; policies return a query and observation-to-action map. Choice values are shuffled deterministically per task. Empty feasible sets are accepted when appropriate.

The full companion `QUESTION_BANK.zh-CN.md` includes every question, every option, every input, and every reference answer. The following family table describes all contracts, and the per-object pages index all 1,400 task instances with four actual website renders per object. No page implies an independent human annotation.

3. Atomic Operations

Identification maps an image-highlighted service module to its ID. Counting groups visible parts by module. Color and position questions read explicit attributes and world bounding-box centers. Joint type, parents, roots, and incident-degree questions inspect the declared attachment graph.

Recoloring changes exactly one part’s color. Addition chooses an exact missing part descriptor. Removal identifies leaf modules. Replacement maximizes stiffness under a cost limit. Translation inverts an observed offset; rotation selects a shortest signed yaw correction. Next-module questions test predecessor completion. Inventory checks conserve piece count; boundary questions identify every incident joint; no-operation questions reward preserving an

already-correct state.

These are controlled contracts, and several public inputs disclose the exact attributes needed for a solution. They are not all image-only perception tasks. One must report the information condition alongside accuracy and run ablations before assigning a visual interpretation.

4. Meta and Physical Tasks

Prefix tasks detect the first violated predecessor. Accessibility reads actual continuous cuboid shape-cast records. Counterfactual tasks remove a specified node and recompute undirected reachability to anchored roots; the explicitly removed node is excluded from the answer. Dynamics asks for the largest sampled translational displacement from eleven actual Rapier measurements.

Kinematic questions use declared joint axes and limits, not empirical actuation success. Diagnosis matches constraint type and observed endpoints, accepting all matching interfaces. Inspection optimizes entropy reduction per unit query cost. Abstention checks agreement of actions across possible worlds. Posterior inference conditions a finite uniform prior on an observation. Pareto enumeration retains all non-dominated cost/mass/stiffness alternatives.

5. Action State Machine

The initial fact set, action catalog, preconditions, forbidden facts, add/delete effects, costs, budget, and goals are public. Assembly/disassembly changes module presence. A service intervention requires support before access, access before removal or modification, replacement before verification, closure before support release. Actions on independent modules may be interleaved. The evaluator never sorts or repairs a submitted program.

For every action, the replay records accepted status, first error, current facts, visible modules, color overrides, and cumulative cost. Reference and submitted model JSON use the same interpreter in the original webpage. A visible explosion is a display transformation only. Neither symbolic acceptance nor rendered motion is claimed to certify clutch forces or robot insertion.

6. Integration

Multi-fault programs combine several targets with one budget and final goal. Scheduling checks integer starts, predecessor finishes, resource exclusion, and makespan. Policies must map all observations to actions correct in every compatible world; success on one sampled branch is insufficient. Unknown actions, repeated completed actions, omitted terminal actions, conflicts, and missing policy branches are tested with negative controls.

7. Simulator and Quality Boundaries

Rapier 0.20.0 runs at 120 Hz and 12 solver iterations. Modules are compound cuboid bodies. Bases are fixed. Holding motors use stiffness 120, damping 20, and maximum force 100 in scene units. Three access casts use a 0.18-unit half-extent probe and exclude the service module. The static query world is independent of settling.

After 120 nominal steps, an impulse (10, 1.8, 3.5) is applied one unit above the target body origin. The next 120 steps produce eleven samples. Nominal drift is evaluated before the impulse. Only 17 of 40 layouts have maximum body-origin drift below 0.35; the remaining layouts are diagnostic records, not certified stable constructions. Changing motor gains, collision geometry, or anchor assumptions changes the experiment and requires a new protocol/version.

8. Pilot and Reproduction

The GPT-4.1 mini pilot contains 16 fixed questions, one per level/layer cell. Families are boundary, information gain, service repair, and scheduling. Fourteen responses succeed at a total cost of \$0.011304. Each cell has one response, so no stable accuracy estimate or model ranking follows. Input hashes, generation IDs, provider receipts, and raw predictions are archived separately from the reference answers.

The 1,400 oracle checks validate the evaluator; 200 public-precondition solutions validate action execution. A constant-choice/empty-answer control passes 268 cases. These controls do not establish task novelty or human difficulty. The figures use original-site canvas captures, and byte-identical model records are shipped to the website and offline release.

From the repository root:

```
npm --prefix benchmark run h2:release
npm run start:local
npm --prefix benchmark run h2:render
npm --prefix benchmark run h2:docs
npm --prefix benchmark test
npm run build
```

Model inference is a separate paid command and is not required for offline score replay. The release manifest binds

data and generator hashes. Public-input files exclude the answer; the review website intentionally includes a reference-answer control.

9. All 35 Task Families

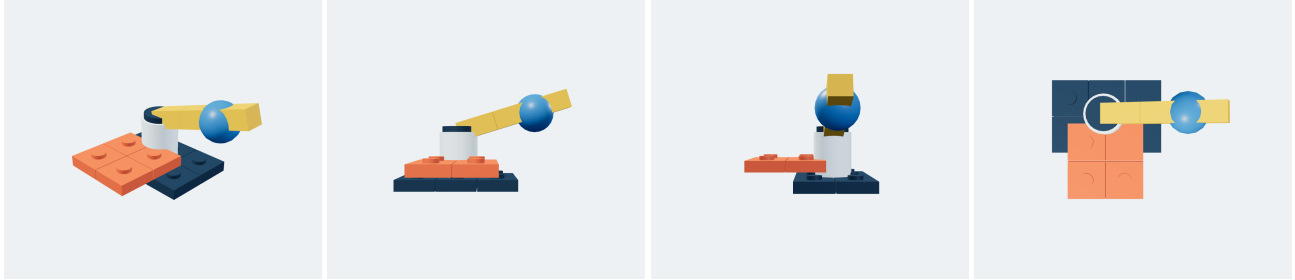
Family	Layer	Output	Evidence
identify	atomic	single-choice	model-state
count	atomic	single-choice	model-state
color	atomic	single-choice	model-state
position	atomic	single-choice	model-state
joint-type	atomic	single-choice	model-state
parent	atomic	single-choice	model-state
anchor	atomic	single-choice	model-state
degree	atomic	single-choice	model-state
recolor	atomic	single-choice	model-state
add	atomic	single-choice	model-state
remove	atomic	single-choice	model-state
replace	atomic	single-choice	model-state
translate	atomic	single-choice	model-state
rotate	atomic	single-choice	model-state
next-module	atomic	single-choice	model-state
inventory	atomic	single-choice	model-state
boundary	atomic	single-choice	model-state
no-op	atomic	single-choice	model-state
prefix	metacognitive	single-choice	support-graph
access	metacognitive	multiple-choice	Rapier
counterfactual	metacognitive	single-choice	support-graph
dynamic	metacognitive	single-choice	Rapier
kinematic	metacognitive	multiple-choice	model-state
diagnosis	metacognitive	multiple-choice	model-state
information-gain	metacognitive	multiple-choice	finite-world
abstention	metacognitive	single-choice	finite-world
posterior	metacognitive	single-choice	finite-world
pareto	metacognitive	multiple-choice	model-state
assembly	procedural	actions	state-machine
disassembly	procedural	actions	state-machine
service-repair	procedural	actions	state-machine
compound-edit	procedural	actions	state-machine
multi-fault	integrative	actions	state-machine
scheduling	integrative	schedule	resource-schedule
policy	integrative	policy	finite-world

Each row appears on ten source layouts at each of four levels. Rows sharing an object remain correlated; source-level reporting and family-grouped confirmation are required. The per-object pages below are the complete release index. A model’s render count is not a source count, and shared part counts do not imply equivalent task difficulty.

10. Complete Object and Question Index

10.1. 1. Camera Gimbal (D1)

16 visible parts, 4 modules, 3 joints. Nominal drift: 0.0123; tolerance outcome: within. Inter-module envelope overlaps: 4. These counts do not certify connector legality.

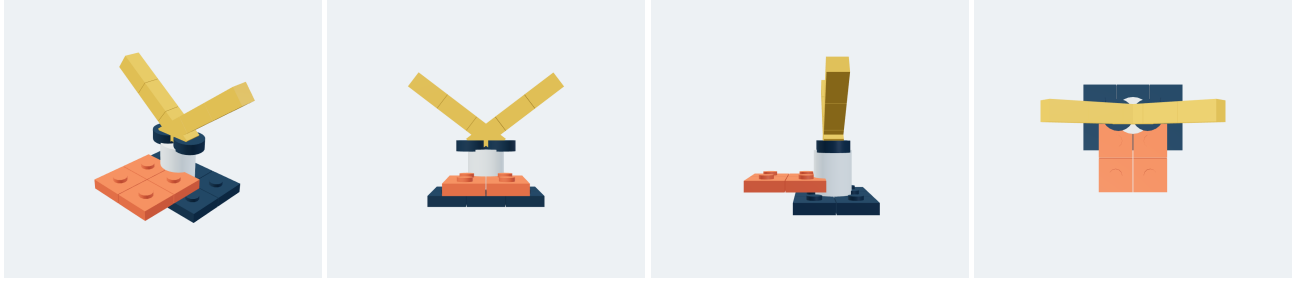


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.2. 2. Folding Gripper (D1)

19 visible parts, 5 modules, 4 joints. Nominal drift: 0.0840; tolerance outcome: within. Inter-module envelope overlaps: 12. These counts do not certify connector legality.

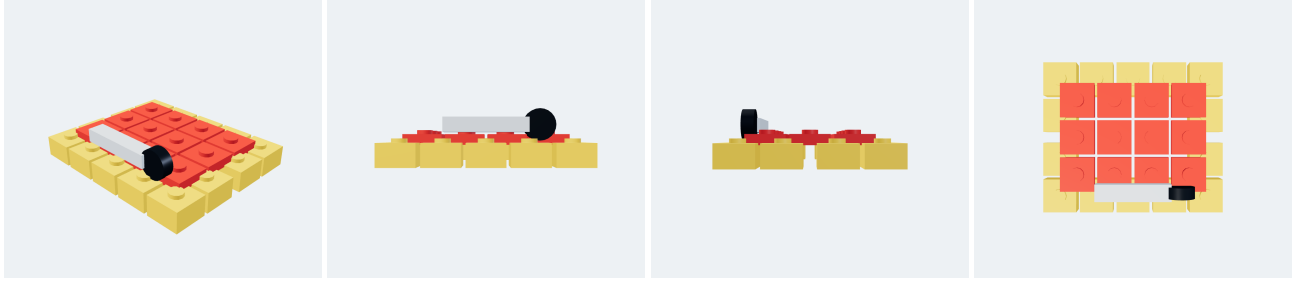


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.3. 3. Hinged Safety Hatch (D1)

28 visible parts, 3 modules, 2 joints. Nominal drift: 0.0304; tolerance outcome: within. Inter-module envelope overlaps: 26. These counts do not certify connector legality.

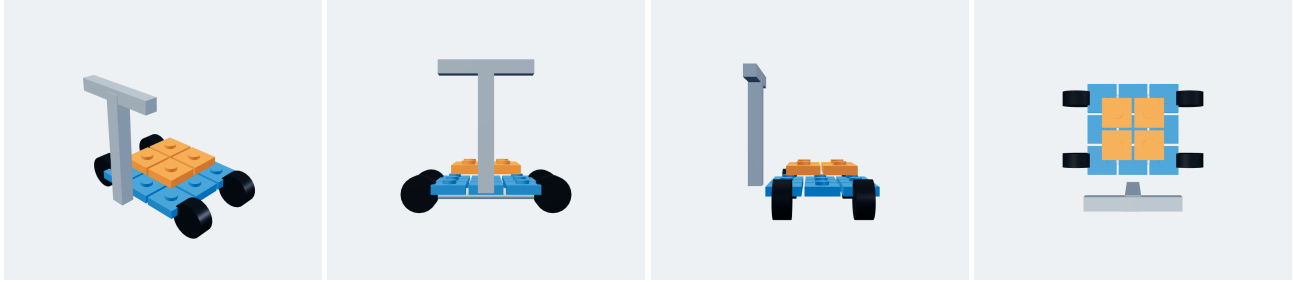


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.4. 4. Maintenance Trolley (D1)

21 visible parts, 5 modules, 4 joints. Nominal drift: 0.0000; tolerance outcome: within. Inter-module envelope overlaps: 10. These counts do not certify connector legality.

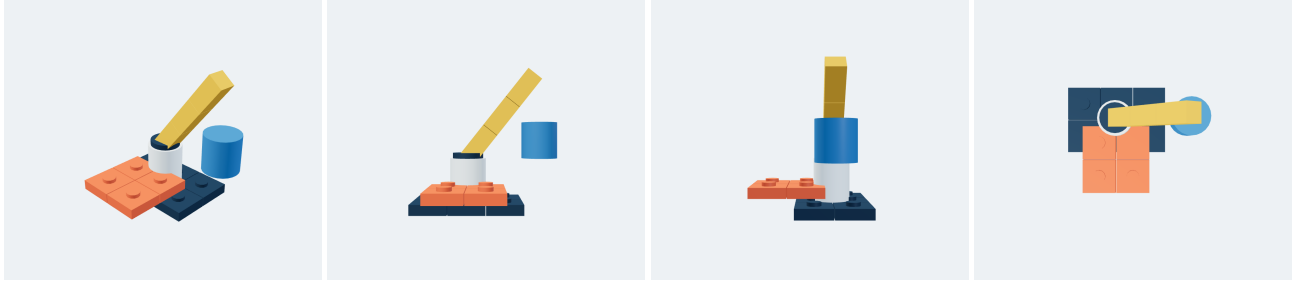


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.5. 5. Ratchet Hoist (D1)

16 visible parts, 4 modules, 3 joints. Nominal drift: 0.0062; tolerance outcome: within. Inter-module envelope overlaps: 4. These counts do not certify connector legality.



The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.6. 6. Sample Carousel (D1)

15 visible parts, 4 modules, 3 joints. Nominal drift: 0.0034; tolerance outcome: within. Inter-module envelope overlaps: 2. These counts do not certify connector legality.

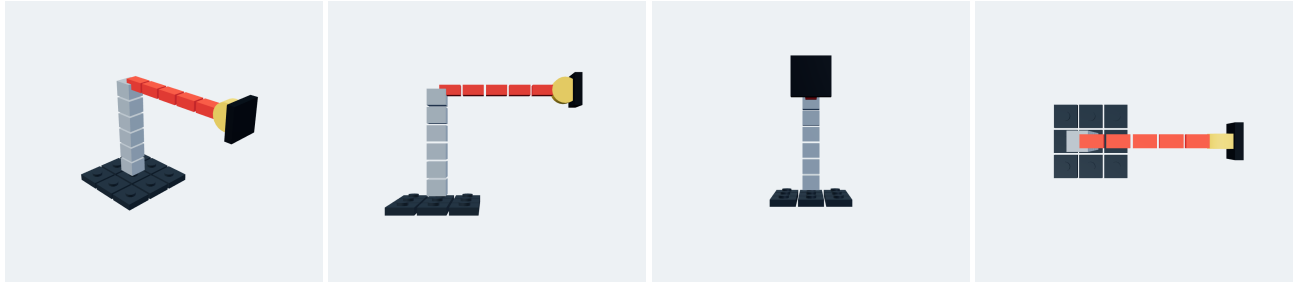


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.7. 7. Signal Switch Stand (D1)

22 visible parts, 4 modules, 3 joints. Nominal drift: 2.0555; tolerance outcome: outside. Inter-module envelope overlaps: 2. These counts do not certify connector legality.

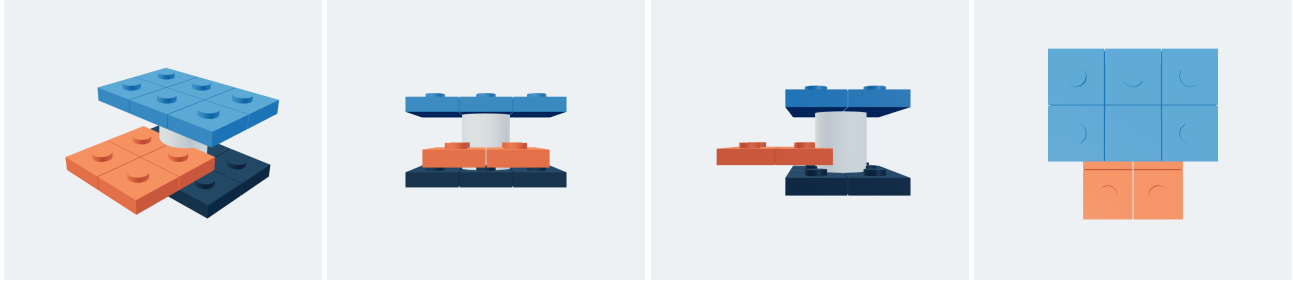


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.8. 8. Rail Tool Drawer (D1)

17 visible parts, 4 modules, 3 joints. Nominal drift: 0.0289; tolerance outcome: within. Inter-module envelope overlaps: 4. These counts do not certify connector legality.

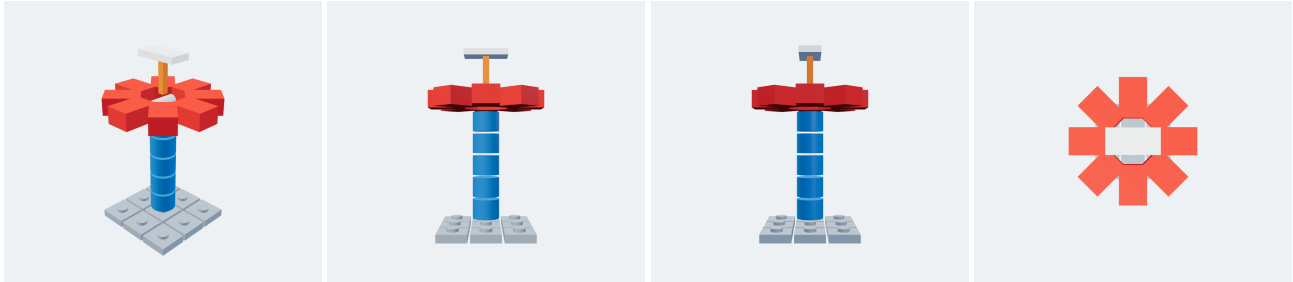


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.9. 9. Valve Control Stand (D1)

25 visible parts, 3 modules, 2 joints. Nominal drift: 0.0000; tolerance outcome: within. Inter-module envelope overlaps: 0. These counts do not certify connector legality.

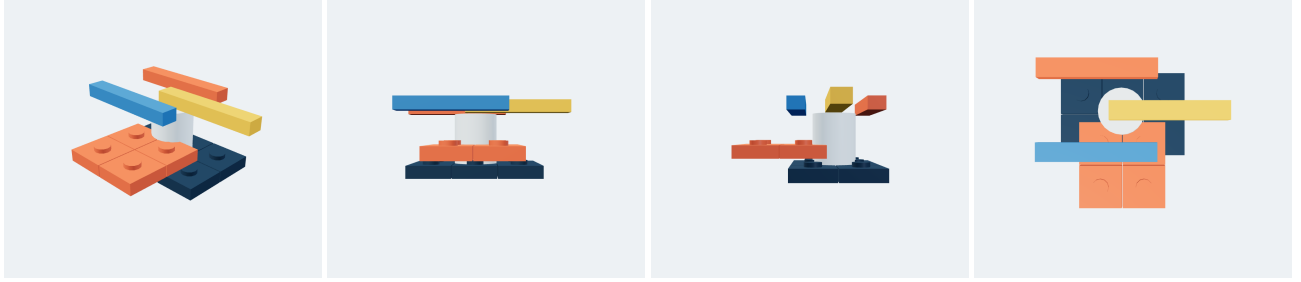


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.10. 10. Wind Vane (D1)

14 visible parts, 4 modules, 3 joints. Nominal drift: 0.0052; tolerance outcome: within. Inter-module envelope overlaps: 2. These counts do not certify connector legality.

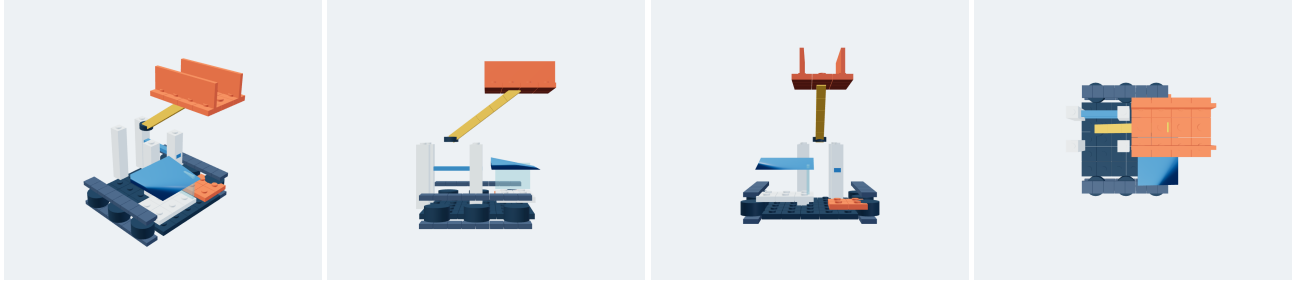


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.11. 11. Articulated Loader (D2)

126 visible parts, 8 modules, 7 joints. Nominal drift: 1.7969; tolerance outcome: outside. Inter-module envelope overlaps: 31. These counts do not certify connector legality.

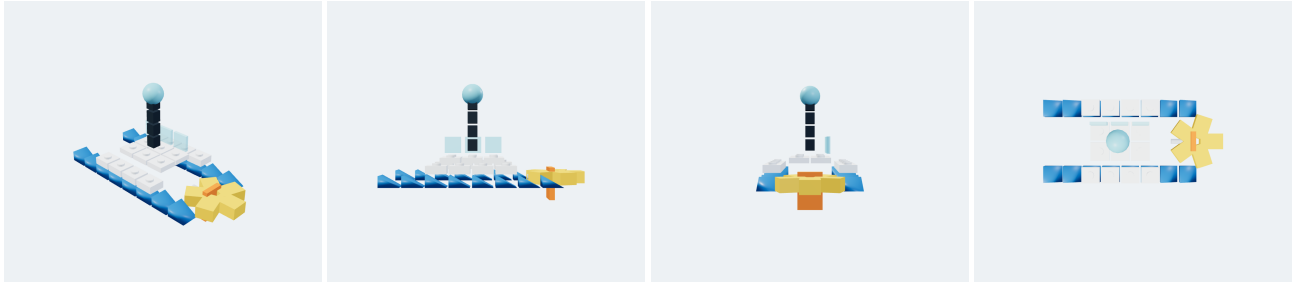


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.12. 12. Canal Inspection Skiff (D2)

45 visible parts, 5 modules, 4 joints. Nominal drift: 5.4774; tolerance outcome: outside. Inter-module envelope overlaps: 7. These counts do not certify connector legality.

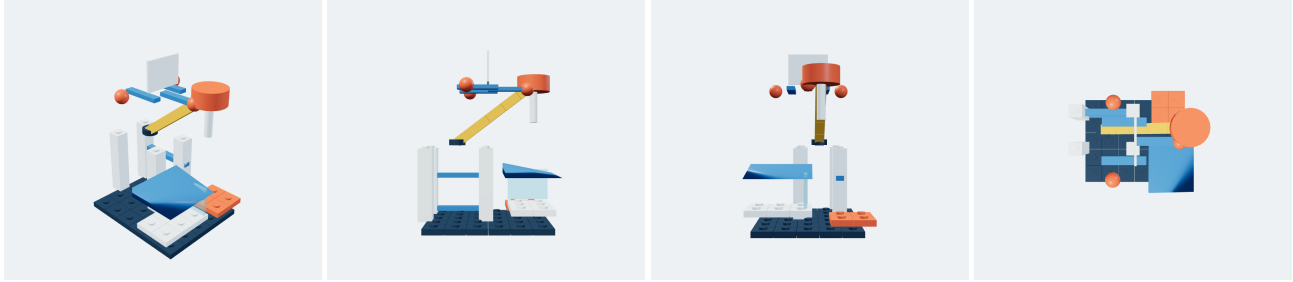


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.13. 13. Folding Weather Mast (D2)

83 visible parts, 7 modules, 6 joints. Nominal drift: 0.7772; tolerance outcome: outside. Inter-module envelope overlaps: 16. These counts do not certify connector legality.

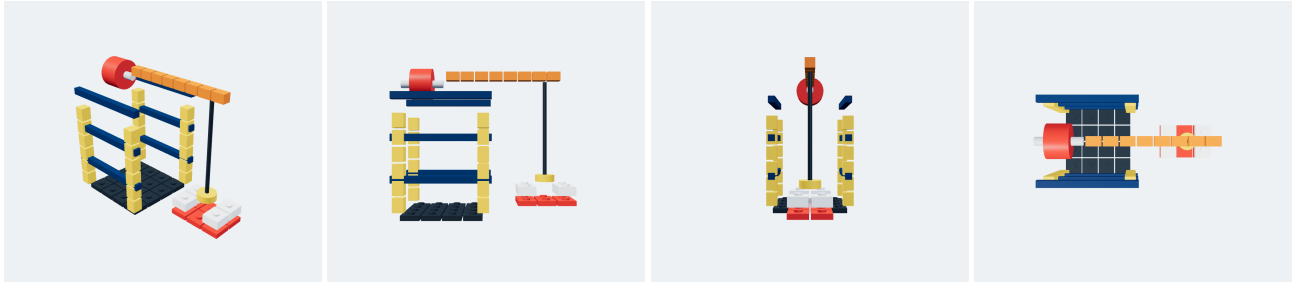


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.14. 14. Rescue Winch Tower (D2)

68 visible parts, 6 modules, 5 joints. Nominal drift: 2.1780; tolerance outcome: outside. Inter-module envelope overlaps: 0. These counts do not certify connector legality.



The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.15. 15. Scissor Service Lift (D2)

108 visible parts, 9 modules, 8 joints. Nominal drift: 10.0052; tolerance outcome: outside. Inter-module envelope overlaps: 26. These counts do not certify connector legality.

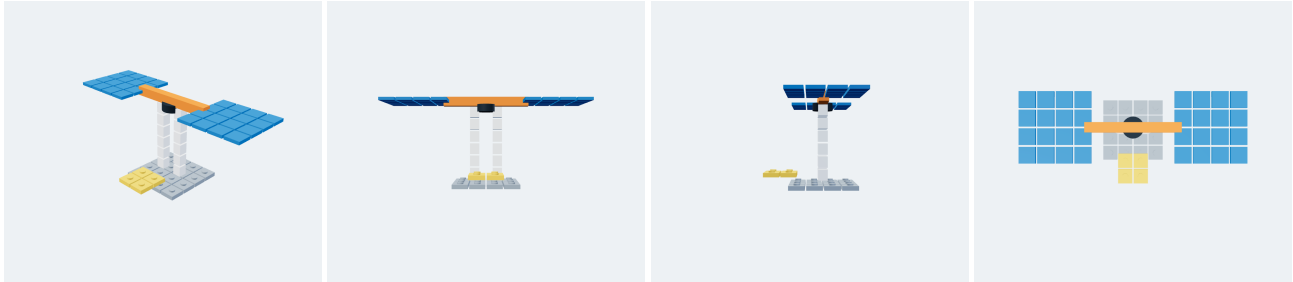


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.16. 16. Solar Tracker Array (D2)

66 visible parts, 6 modules, 5 joints. Nominal drift: 0.0000; tolerance outcome: within. Inter-module envelope overlaps: 6. These counts do not certify connector legality.

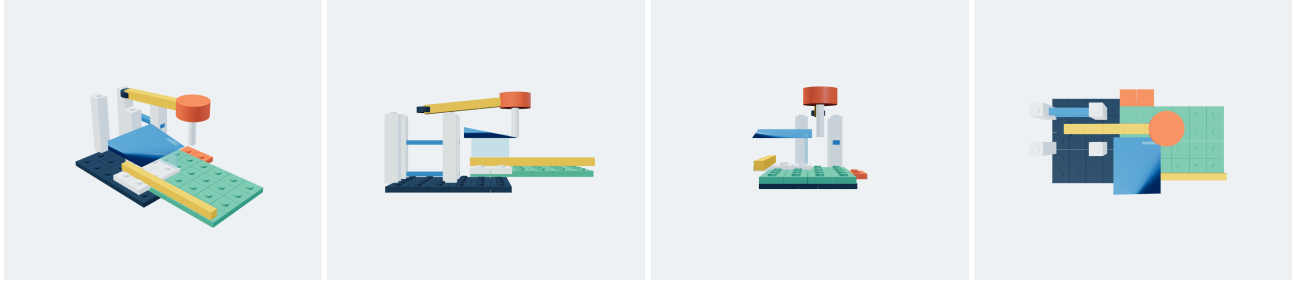


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.17. 17. Swing Loading Dock (D2)

101 visible parts, 7 modules, 6 joints. Nominal drift: 1.0934; tolerance outcome: outside. Inter-module envelope overlaps: 26. These counts do not certify connector legality.

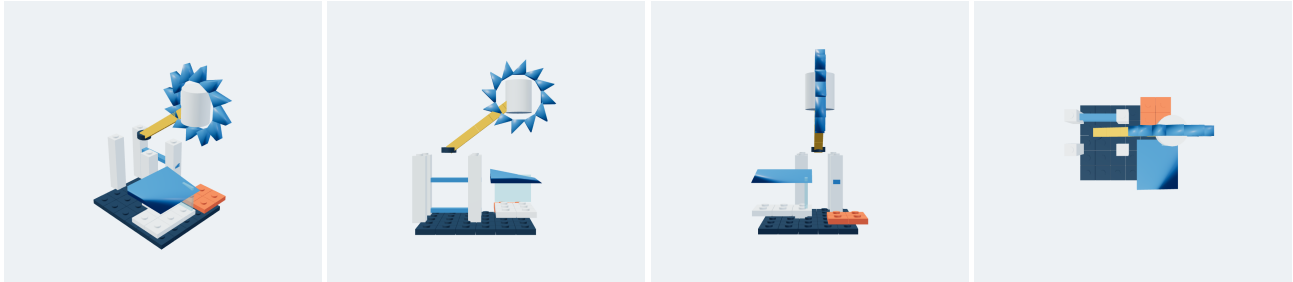


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.18. 18. Telescopic Optics Cart (D2)

87 visible parts, 6 modules, 5 joints. Nominal drift: 1.7592; tolerance outcome: outside. Inter-module envelope overlaps: 17. These counts do not certify connector legality.

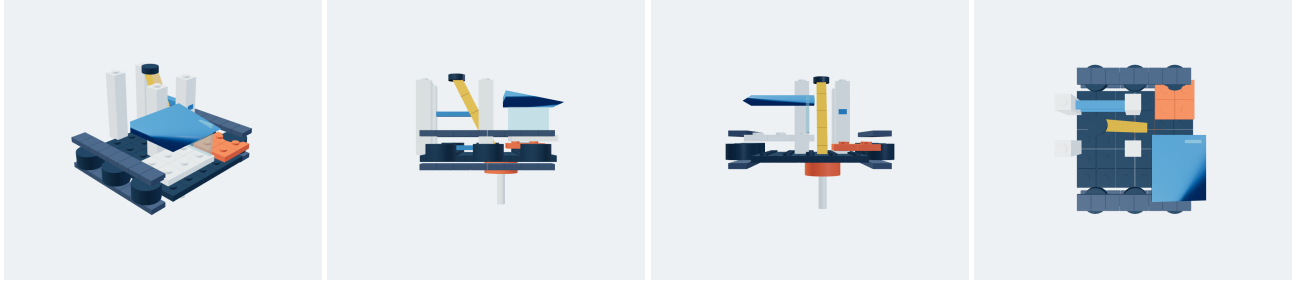


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.19. 19. Tracked Core Drill (D2)

114 visible parts, 8 modules, 7 joints. Nominal drift: 0.2291; tolerance outcome: within. Inter-module envelope overlaps: 39. These counts do not certify connector legality.

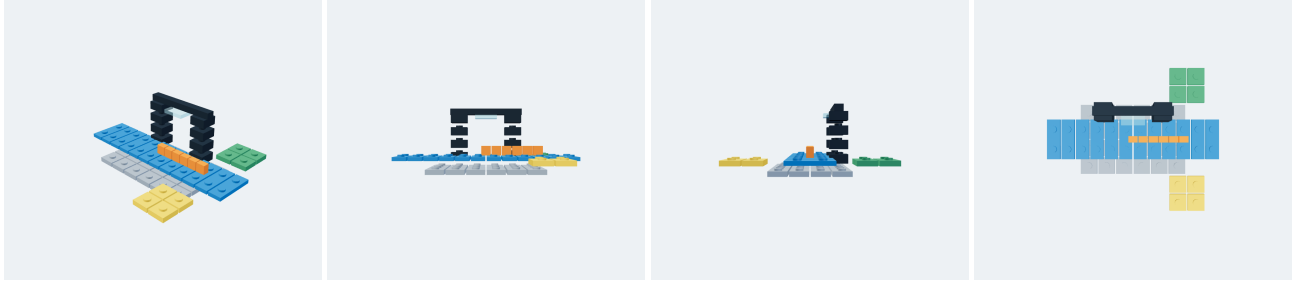


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.20. 20. Warehouse Sorter (D2)

60 visible parts, 6 modules, 5 joints. Nominal drift: 0.0014; tolerance outcome: within. Inter-module envelope overlaps: 4. These counts do not certify connector legality.

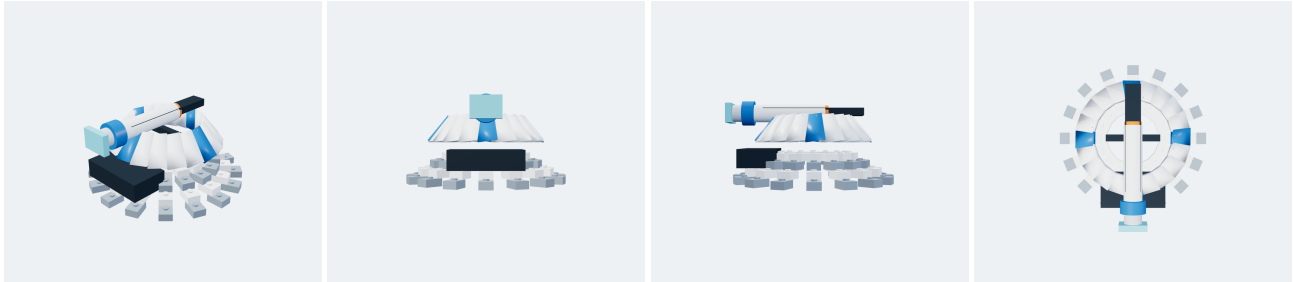


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.21. 21. Adaptive Radio Observatory (D3)

92 visible parts, 4 modules, 3 joints. Nominal drift: 1.3771; tolerance outcome: outside. Inter-module envelope overlaps: 7. These counts do not certify connector legality.

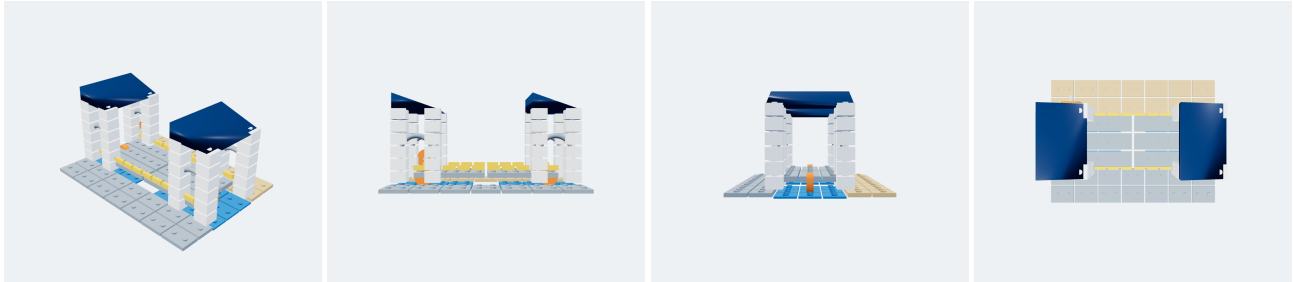


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.22. 22. Bascule Canal Gate (D3)

142 visible parts, 6 modules, 7 joints. Nominal drift: 0.1000; tolerance outcome: within. Inter-module envelope overlaps: 28. These counts do not certify connector legality.

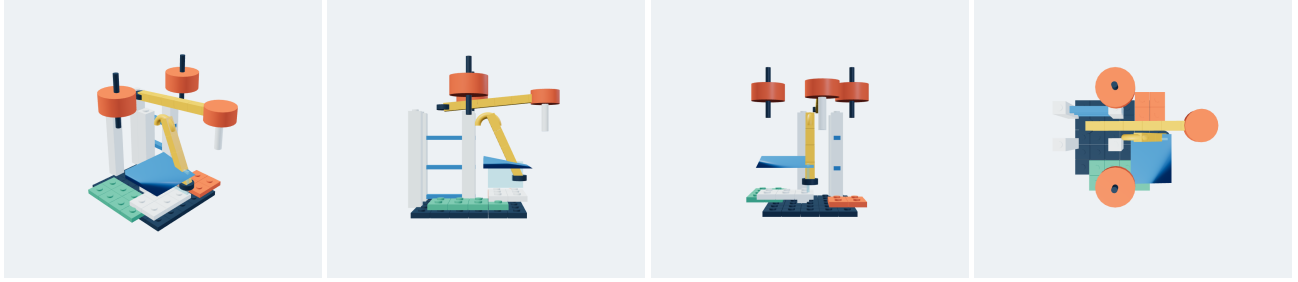


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.23. 23. Cable Inspection Robot (D3)

108 visible parts, 11 modules, 10 joints. Nominal drift: 1.0669; tolerance outcome: outside. Inter-module envelope overlaps: 24. These counts do not certify connector legality.

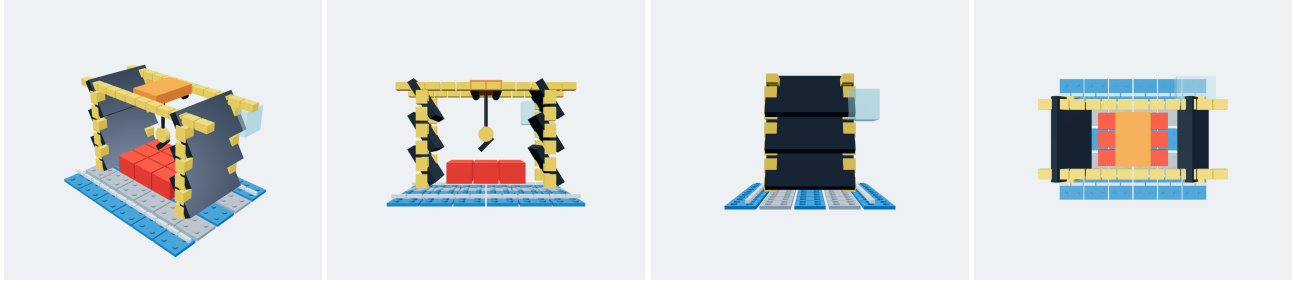


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.24. 24. Harbor Container Crane (D3)

126 visible parts, 5 modules, 5 joints. Nominal drift: 0.8147; tolerance outcome: outside. Inter-module envelope overlaps: 1. These counts do not certify connector legality.

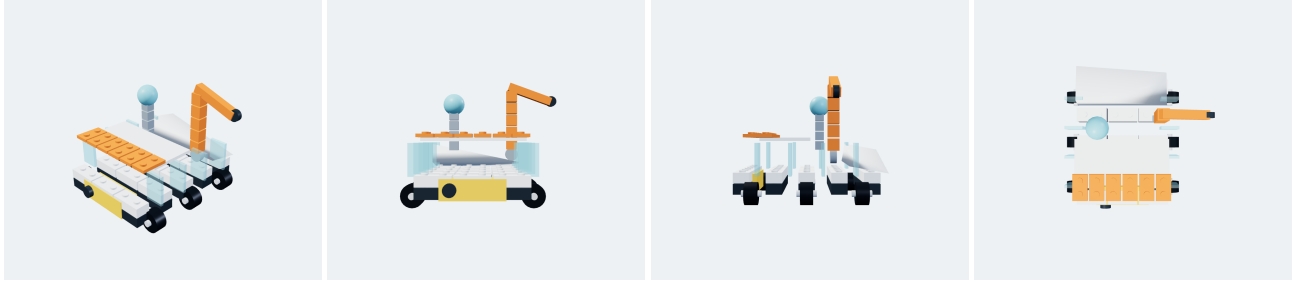


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.25. 25. Orbital Service Rover (D3)

73 visible parts, 10 modules, 9 joints. Nominal drift: 0.0000; tolerance outcome: within. Inter-module envelope overlaps: 18. These counts do not certify connector legality.

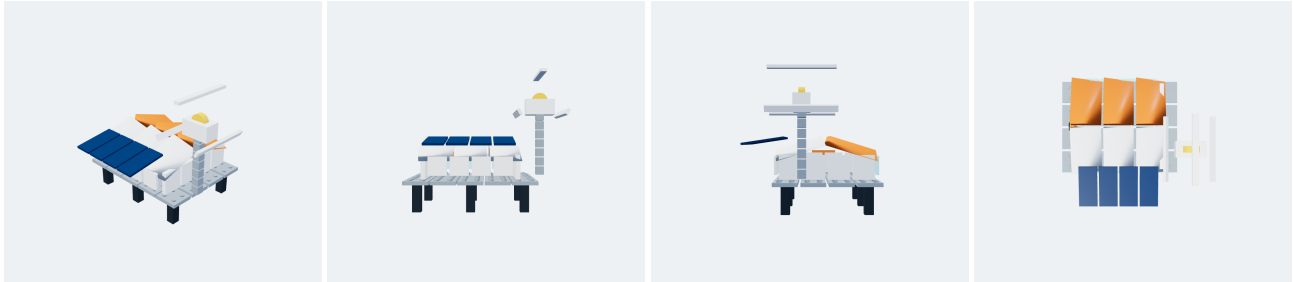


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.26. 26. Polar Research Station (D3)

66 visible parts, 5 modules, 4 joints. Nominal drift: 0.0000; tolerance outcome: within. Inter-module envelope overlaps: 11. These counts do not certify connector legality.

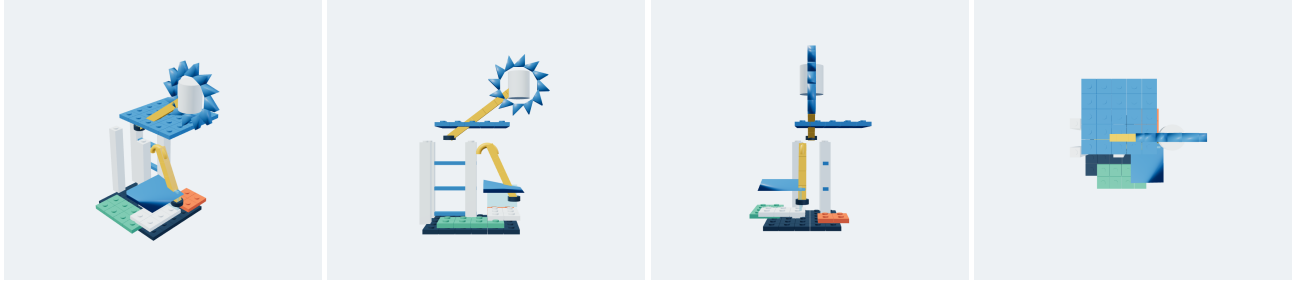


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.27. 27. Retractable Array Radar (D3)

140 visible parts, 10 modules, 9 joints. Nominal drift: 2.4487; tolerance outcome: outside. Inter-module envelope overlaps: 31. These counts do not certify connector legality.



The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.28. 28. Rotary Bucket Excavator (D3)

154 visible parts, 11 modules, 10 joints. Nominal drift: 4.8438; tolerance outcome: outside. Inter-module envelope overlaps: 49. These counts do not certify connector legality.

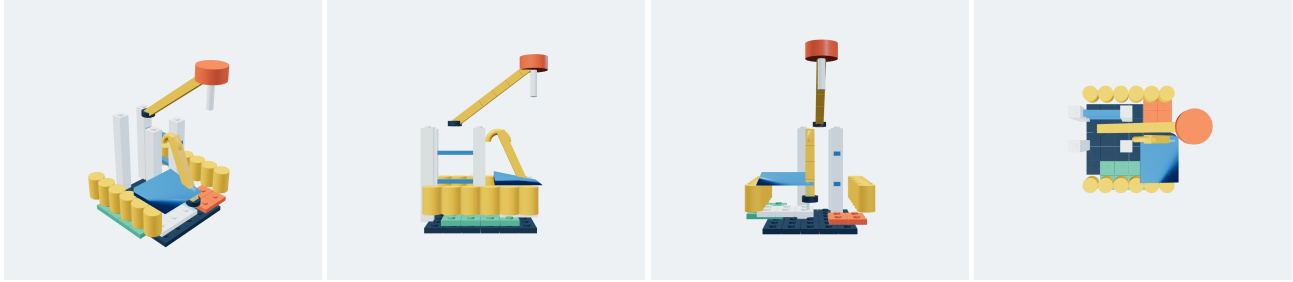


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.29. 29. Subsea Manipulator (D3)

116 visible parts, 11 modules, 10 joints. Nominal drift: 2.1669; tolerance outcome: outside. Inter-module envelope overlaps: 24. These counts do not certify connector legality.

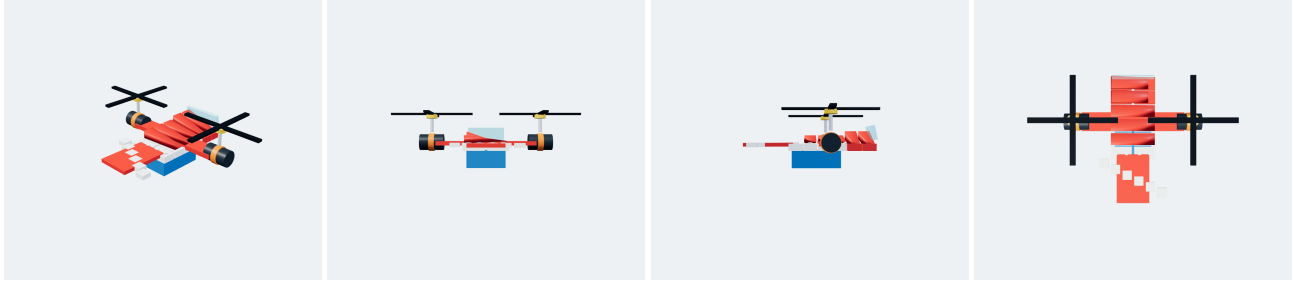


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.30. 30. Wildfire Tiltrotor (D3)

43 visible parts, 6 modules, 5 joints. Nominal drift: 0.0000; tolerance outcome: within. Inter-module envelope overlaps: 10. These counts do not certify connector legality.

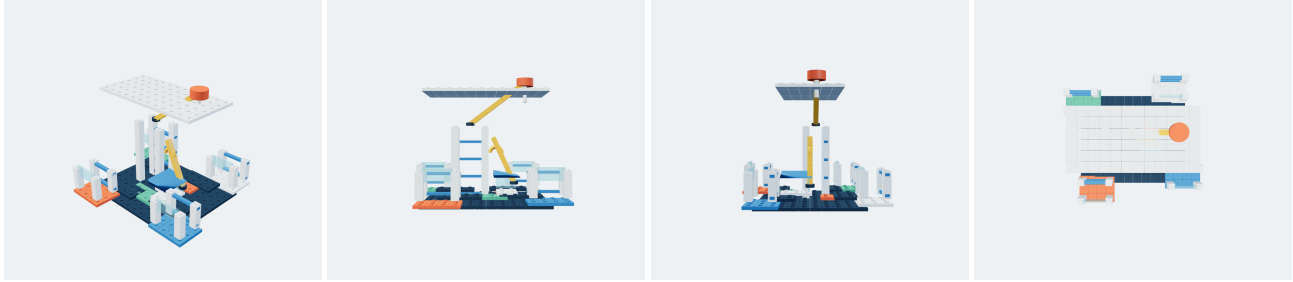


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.31. 31. Aircraft Service Hangar (D4)

437 visible parts, 22 modules, 21 joints. Nominal drift: 0.1790; tolerance outcome: within. Inter-module envelope overlaps: 55. These counts do not certify connector legality.

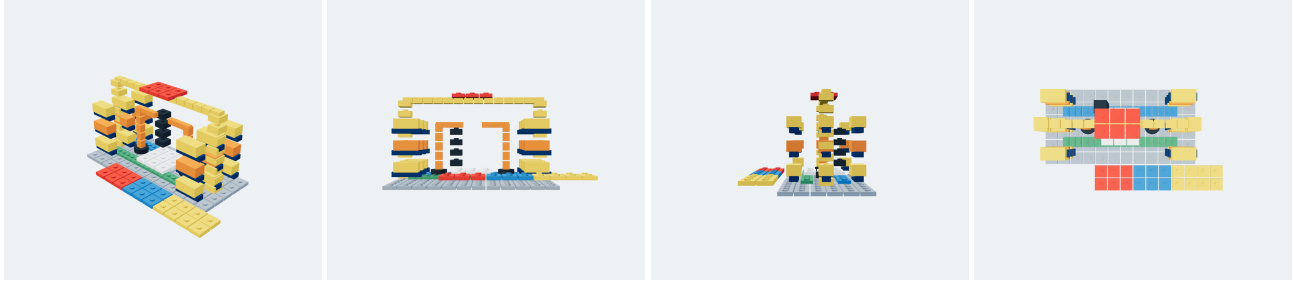


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.32. 32. Automated Cargo Terminal (D4)

231 visible parts, 13 modules, 12 joints. Nominal drift: 17.6735; tolerance outcome: outside. Inter-module envelope overlaps: 28. These counts do not certify connector legality.

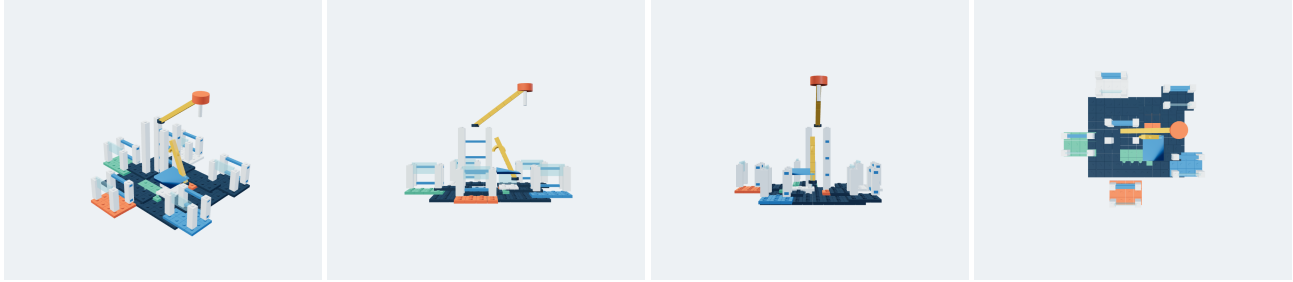


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.33. 33. Automated Recycling Plant (D4)

404 visible parts, 24 modules, 23 joints. Nominal drift: 2.3564; tolerance outcome: outside. Inter-module envelope overlaps: 59. These counts do not certify connector legality.

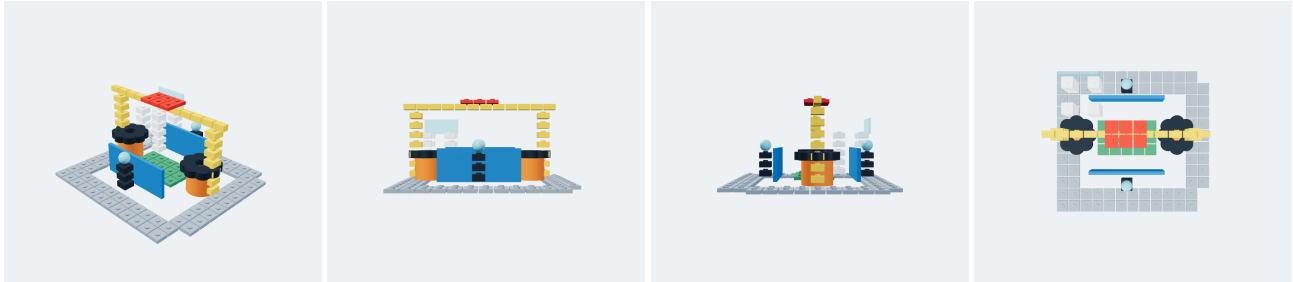


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.34. 34. Flood Response Lock (D4)

214 visible parts, 11 modules, 10 joints. Nominal drift: 4.2473; tolerance outcome: outside. Inter-module envelope overlaps: 10. These counts do not certify connector legality.

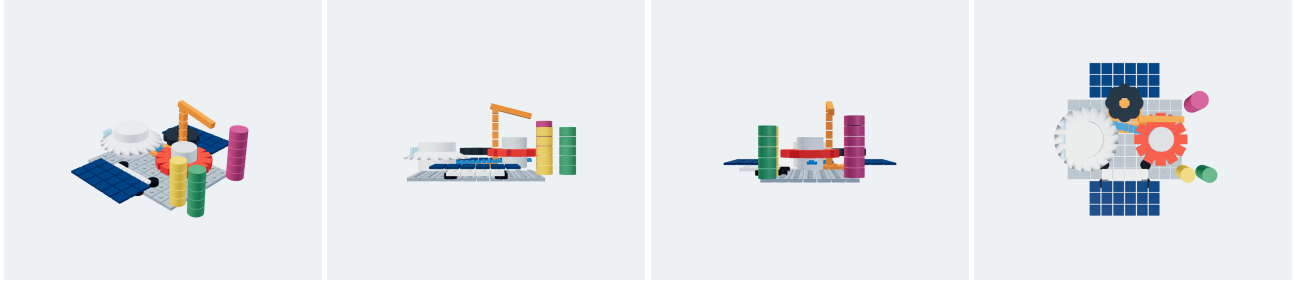


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.35. 35. Lunar Sample Refinery (D4)

218 visible parts, 12 modules, 11 joints. Nominal drift: 0.4023; tolerance outcome: outside. Inter-module envelope overlaps: 15. These counts do not certify connector legality.

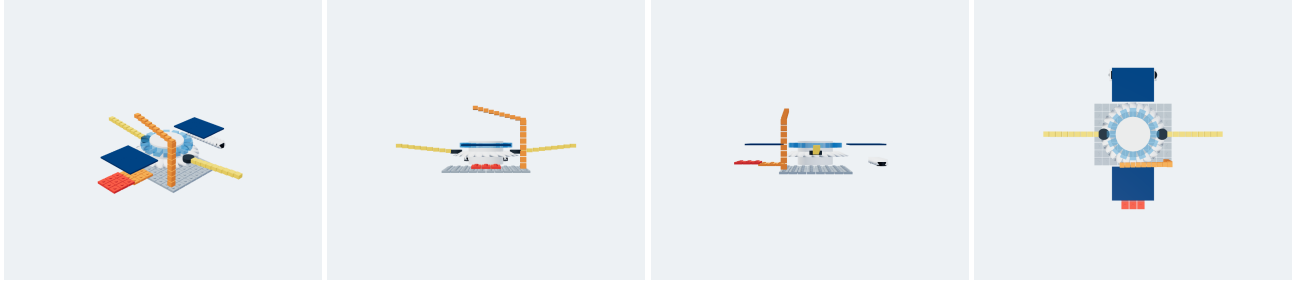


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.36. 36. Orbital Docking Yard (D4)

233 visible parts, 11 modules, 10 joints. Nominal drift: 2.1352; tolerance outcome: outside. Inter-module envelope overlaps: 12. These counts do not certify connector legality.

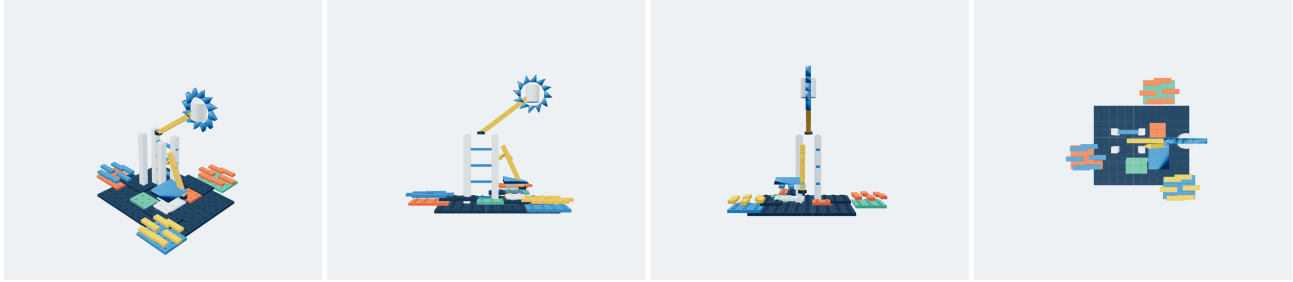


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.37. 37. Orbital Telescope Yard (D4)

298 visible parts, 18 modules, 17 joints. Nominal drift: 2.1919; tolerance outcome: outside. Inter-module envelope overlaps: 45. These counts do not certify connector legality.

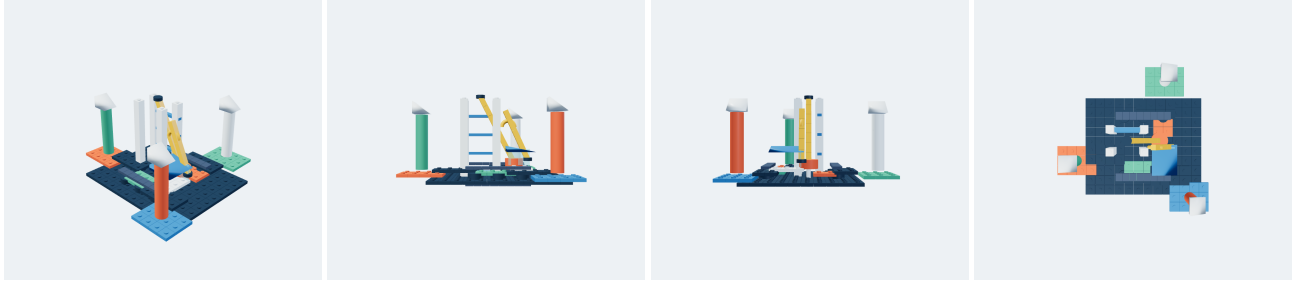


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.38. 38. Polar Drilling Complex (D4)

331 visible parts, 20 modules, 19 joints. Nominal drift: 5.8032; tolerance outcome: outside. Inter-module envelope overlaps: 94. These counts do not certify connector legality.

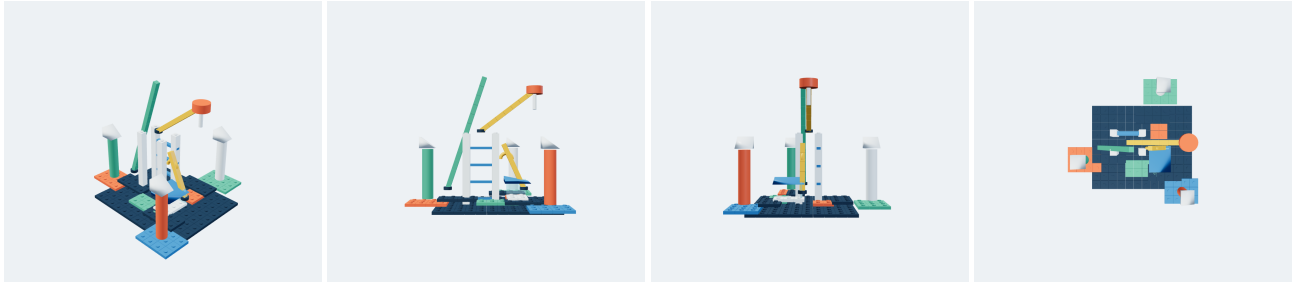


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.39. 39. Rescue Launch Complex (D4)

306 visible parts, 19 modules, 18 joints. Nominal drift: 1.9827; tolerance outcome: outside. Inter-module envelope overlaps: 47. These counts do not certify connector legality.

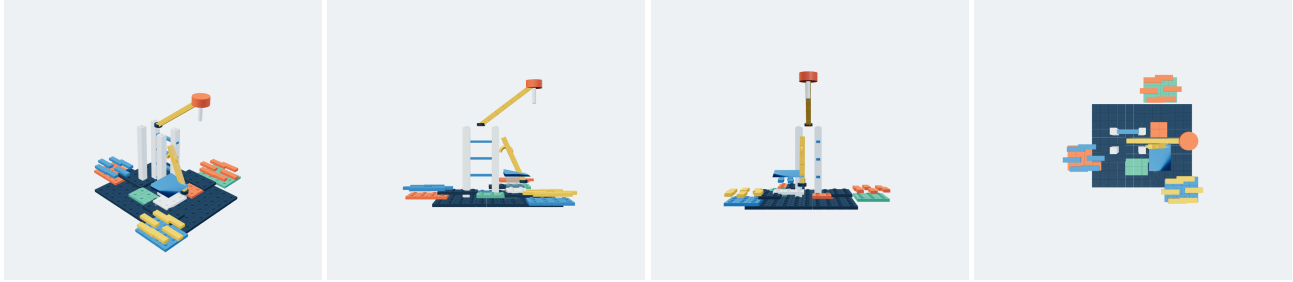


The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world

10.40. 40. Tidal Energy Station (D4)

287 visible parts, 18 modules, 17 joints. Nominal drift: 2.3616; tolerance outcome: outside. Inter-module envelope overlaps: 42. These counts do not certify connector legality.



The following 35 tasks share this source object. Full inputs, choices and answers are in the companion question bank.

No.	Family	Layer	Output	Evidence
1	identify	atomic	single-choice	model-state
2	count	atomic	single-choice	model-state
3	color	atomic	single-choice	model-state
4	position	atomic	single-choice	model-state
5	joint-type	atomic	single-choice	model-state
6	parent	atomic	single-choice	model-state
7	anchor	atomic	single-choice	model-state
8	degree	atomic	single-choice	model-state
9	recolor	atomic	single-choice	model-state
10	add	atomic	single-choice	model-state
11	remove	atomic	single-choice	model-state
12	replace	atomic	single-choice	model-state
13	translate	atomic	single-choice	model-state
14	rotate	atomic	single-choice	model-state
15	next-module	atomic	single-choice	model-state
16	inventory	atomic	single-choice	model-state
17	boundary	atomic	single-choice	model-state
18	no-op	atomic	single-choice	model-state
19	prefix	metacognitive	single-choice	support-graph
20	access	metacognitive	multiple-choice	Rapier
21	counterfactual	metacognitive	single-choice	support-graph
22	dynamic	metacognitive	single-choice	Rapier
23	kinematic	metacognitive	multiple-choice	model-state
24	diagnosis	metacognitive	multiple-choice	model-state
25	information-gain	metacognitive	multiple-choice	finite-world
26	abstention	metacognitive	single-choice	finite-world
27	posterior	metacognitive	single-choice	finite-world
28	pareto	metacognitive	multiple-choice	model-state
29	assembly	procedural	actions	state-machine
30	disassembly	procedural	actions	state-machine
31	service-repair	procedural	actions	state-machine
32	compound-edit	procedural	actions	state-machine
33	multi-fault	integrative	actions	state-machine
34	scheduling	integrative	schedule	resource-schedule
35	policy	integrative	policy	finite-world